pieces seem to have been a few kilometers across. Their impacts with Jupiter were spectacular.

No one knew beforehand what these multiple impacts into the atmosphere and clouds of Jupiter would do. Perhaps the cometary fragments, surrounded by halos of dust, were much smaller than they seemed. Or perhaps they were not coherent bodies at all, but loosely consolidated—something like a heap of gravel with all the particles traveling through space together, in nearly identical orbits. If either of these possibilities were true, Jupiter might swallow the comets without a trace. Other astronomers thought there would at least be bright fireballs and giant plumes as the cometary fragments plunged into the atmosphere. Still others suggested that the dense cloud of fine particles accompanying the fragments of Comet Shoemaker-Levy 9 into Jupiter would disrupt the magnetosphere of Jupiter or form a new ring.

A comet this size should impact Jupiter, it is calculated, only once every thousand years. It's the astronomical event not of one lifetime, but of a dozen. Nothing on this scale has occurred since the invention of the telescope. So in mid-July 1994, in a beautifully coordinated international scientific effort, telescopes all over the Earth and in space turned towards Jupiter.

Astronomers had over a year to prepare. The trajectories of the fragments in their orbits around Jupiter were estimated. It was discovered that they would all hit Jupiter. Predictions of the timing were refined. Disappointingly, the calculations revealed that all impacts would occur on the night side of Jupiter, the side invisible from the Earth (although accessible to the *Galileo* and *Voyager* spacecraft in the outer Solar System). But, happily, all impacts would occur only a few minutes before the Jovian dawn, before the impact site would be carried by Jupiter's rotation into the line of sight from Earth.

The appointed moment for the impact of the first piece, Fragment A, came and went. There were no reports from ground-based telescopes. Planetary scientists stared with increasing gloom at a television monitor displaying the data transmitted to the Space Telescope Science Institute in Baltimore from the Hubble Space